Capital Budgeting Techniques

Presented By

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Slide 1: Capital Budgeting

"Capital budgeting is a process used by organizations to evaluate and select long-term investment projects that will create value for the company."

Capital Budgeting is defined as the firm's formal process for the acquisition and investment of capital"

"Capital Budgeting is long-term planning for making and financing proposed capital outlays." Charles T Horngreen

Slide 2:The Main features of Capital budgeting are

i) Potentially large anticipated benefits

ii) a relatively high degree of risk

iii) Relatively long time period between the initial outlays and the anticipated return. **Oster Young**

Slide 3: Significance of capital budgeting:

i)The success and failure of a business mainly depend on how the available resources are being utilized.

ii) Main tool of financial management

iii) All types of capital budgeting decisions are exposed to risk and uncertainty.

iv) They are irreversible in nature.

v) Capital rationing gives sufficient scope for the financial manager to evaluate different proposals and only viable projects must be taken up for investments.

vi) Capital budgeting offers effective control of the cost of capital expenditure projects.

vii)It helps the management to avoid over-investment and under-investments.

Slide 4: Techniques of Capital Budgeting:

Capital budgeting techniques are a set of financial evaluation methods used by businesses and organizations to assess potential investment projects. Common capital budgeting techniques include:

1. Traditional Methods (Non-Discounting Techniques)

i. Pay-back period Method

ii. Accounting Rate of Return Method

2. Modern Methods (Discounting Techniques)

i. Net Present Value Method (NPV)

ii. Profitability Index method (PI)

iii. Internal Rate of Return Method (IRR)

v. Discounted Pay-back Period Method

Slide 5: Payback Period Method

Payback Period Method: The payback period is the required number of years to recover the initial investment. It is based on the concept of cash flow and is a non-discounting technique.

Decision Criteria:

I.If there is only one project in consideration, it would be selected only if it has a payback period as per management's expectation.

ii. In the case of more than one project, a project with a lower payback period should be selected.

Slide 6: Formula for Calculating Payback Period

1. When Cash inflows are even/equal

When the cash inflow of all years is equal, we use the following formula

Payback period = Initial Investment/ Annual Cash Inflow

2. When cash inflows are uneven

When cash inflows of each year are different, we use the formula below

Payback Period = E + B/ C

Where,

E = Year immediately Preceding to year of recovery

B = Amount left to be recovered

C = Cash inflow during the year of final recovery

Note: Before using these values, we must find cumulative cash inflows

Slide 7: Merits of Payback Period Method

i) It is easy to calculate and simple to understand.

ii) It Measures the liquidity of the investment.

iii) This method is handy for industries with uncertainty because it focuses on quickly getting your investment back.

iii) For firms seeking rapid investment recovery, the payback period method is ideal for determining how quickly they can recoup their initial investments.

Slide 8: Demerits of the Payback Period Method

I. This method neglects cash flows occurring after the payback period: This method does not consider the amount of profit earned after the recovery of the cost of investment. Some projects may have higher cash inflows after the payback period.

ii. This method does not consider the time value of money.

iii.This method does not consider the risk associated with the project.

iv. Does not consider profitability of economic life of the project.

Slide 9: Average Rate of Return Method:

The Average Rate of Return (ARR) method, also known as the Accounting Rate of Return (ARR), is a capital budgeting technique used to evaluate the profitability of an investment project.

The formula for Average Rate of Return Method:

Average Rate of Return = (Annual Average Profit)/(Initial Investment) *100%

Decision Criteria:

I) In the case of many projects, a project with a higher ARR or NOI will be selected.

II) In the case of only one project, it would be selected if it earns more than the company's predetermined required rate of return.

Slide 10:Pros & Cons of Accounting Rate of Return Method

Advantage of Accounting Rate of Return:

i. It is simple and easy to calculate.

ii. It takes into account all the savings over the entire period of investment.

iii. It is based on accounting profit rather than cash inflow. Accounting profit can be easily obtained from financial statements.

iv. It measures the benefit in percentage which makes it easier to compare with other projects.

Disadvantages of Accounting Rate of Return Method:

i. This method *ignores the time value of money.*

ii. Like the Payback period, this method also *ignores risk factors.*

iii. This method ignores the size of the investment. Sometimes ARR may be the same for different projects but some of them may involve huge cash flows.

iv. It does not consider the *length of life of the projects.*

Slide 11: Net Present Value Method:

I. The NPV method is a discounted cash flow technique.
II. This method compares cash inflows and cash outflows occurring at different periods.
III. The major characteristic of this method is that it takes into account the time value of money and all cash inflows and outflows are converted to present value.

NPV = Present value of all cash inflows **–** Present value of cash outflow

Decision Criteria:

1. If there is only one project under consideration:

   If NPV is positive …… Accept
   If NPV is negative …….. Reject
   If NPV is zero ……Indifferent

2. If there are more than one project, the project with a higher NPV should be selected

Slide 12: Merits and Demerits of Net Present Value Method

Merits of Net Present Value Method:

i. This method recognizes the *time value of money*. Cash inflows arising at different time intervals are discounted to present values.

iI. This method recognizes the risk involved in the project with the help of discounting rate.

iii. This method is *best for mutually exclusive projects* where only one project is to be selected among many.

iv. This method is considered *best for the wealth maximization* of shareholders as it is based on cash inflows rather than accounting profit.

Demerits of Net Present Value Method:

i. It requires ***difficult calculations***

ii. The NPV technique requires the predetermination of the required rate of return, which itself is a difficult job. If that rate is not correctly taken, then the whole exercise of NPV may give the wrong result.

iii. It does not provide a measure of the project's rate of return, rather it evaluates a proposal against an external variable i.e. minimum rate of return.

IV.This method may not provide satisfactory results in case of projects having different amount of investment and different economic life.

Slide 13: Profitability Index Method (PI)

This method is also known as the Benefit-cost ratio method. It is based on the net present value method and calculates the benefit per dollar investment.

Profitability Index (PI)= (Present value of all cash inflow/ Initial cash outlays)

Decision Criteria:

1. If there is only one project:

If PI is greater than 1……. Accept
If PI is less than 1 ……. Reject
If PI is equal to 1 …….. Indifferent

2. If there are more than one project, the project with higher PI should be selected.

Slide 14: Merits and Demerits of Profitability Index Method:

Merits of Profitability Index Method:

i) The PI considers the time value of money, making it suitable for long-term investments.

i) It measures the value generated per unit of investment, aiding in project prioritization with limited resources.

iii) Particularly useful for ranking projects when budget constraints exist.

iv) PI techniques gives better results in case of projects having different outlays.

v) The PI considers cash flows from project inception to completion, ensuring a thorough assessment of financial viability.

Demerits of Profitability Index

i) PI method Ignores project size that is it may favor smaller projects with higher relative returns, potentially neglecting larger, more impactful investments.

ii) PI method doesn't consider reinvestment rates. It assumes reinvestment at the project's rate, which may not reflect actual market rates, leading to inaccuracies.

iii) PI Method May not be useful for mutually exclusive projects: When choosing between projects, PI can be misleading, as it doesn't account for differing project sizes or durations.

Discounted Payback Period Method

The discounted payback period (DPP) is a capital budgeting technique used to evaluate the profitability of a project or investment. It takes into account the time value of money by discounting future cash flows to their present value. The DPP is calculated as the number of years it takes for the cumulative present value of a project's cash flows to equal the initial investment.

Accept/Reject Criterion of Discounted Payback Period:

Acceptance Criterion:

If the discounted payback period of a project is less than or equal to the predetermined maximum payback period (or threshold), the project is accepted.

Rejection Criterion:

If the discounted payback period exceeds the predetermined maximum payback period, the project is rejected.

Slide 16: Pros and Cons of Discounted Payback Period

Advantages of Discounted Payback Period:

i) Accounts for the time value of money, providing a more accurate measure.

ii) imple and intuitive concept for decision-makers.

iii) Offers a sense of risk by measuring the speed of capital recovery.

iv) Useful for evaluating liquidity, as shorter payback periods mean quicker access to funds.

Disadvantages of Discounted Payback Period:

i) Does not consider cash flows occurring after the payback period.

ii) Requires the choice of a discount rate, which can be subjective.

iv) Fails to assess overall profitability, unlike NPV or IRR.

iv) Treats cash flows before and after the payback period differently.

Tends to favor shorter-term projects, potentially discouraging long-term investments.

The End

Thank You